STAT/MA 41600

In-Class Problem Set #25: October 16, 2026

1. Suppose that X and Y are continuous random variables whose joint probability density function $f_{X,Y}(x,y)$ is constant on the interior of the triangle with vertices at $(0,0)$, $(8,0)$, and $(0,5)$, and otherwise $f_{X,Y}(x,y) = 0$.

1a. Find $P(X < 4)$.

1b. Find $P(X > 2)$.

2. Suppose that X and Y have joint probability density function $f_{X,Y}(x,y) = 14e^{-2x-7y}$ for $x > 0$ and $y > 0$, and otherwise $f_{X,Y}(x,y) = 0$.

2a. Find $P(1 \leq \min(X,Y))$.

2b. Find $P(3 \leq \min(X,Y))$.

2c. Find $P(10 \leq \min(X,Y))$.

2d. Can you generalize? If $a > 0$ is constant, find $P(a \leq \min(X,Y))$.

3a. For the setup in Question 2, find $P(X + Y \geq 2)$.

3b. For the setup in Question 2, find $P(X > 4Y)$.

4. Suppose that, for $0 < x < 5$ and $0 < y < 2$, we have $f_{X,Y}(x,y) = k(5-x)(2-y)$ for some constant k, and otherwise $f_{X,Y}(x,y) = 0$.

4a. What value of k makes $f_{X,Y}$ be a valid probability density function?

4b. What is the probability density function of X?

4c. What is the probability density function of Y?

5. Now consider two random variables X and Y and a constant k, where the joint probability density function of X, Y is $f_{X,Y}(x,y) = ke^{-2x-7y}$ for $y > x > 0$, and otherwise $f_{X,Y}(x,y) = 0$.

5a. What value of k makes $f_{X,Y}$ be a valid probability density function?

5b. What is the probability density function of X?